

SF2524
Matrix computations for large-scale systems
Homework 1

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Problem 1

Let

$$A_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 9 \end{pmatrix} \quad A_2 = \begin{pmatrix} 1 & 2 & 4 \\ 2 & 0 & 2 \\ 3 & 2 & 9 \end{pmatrix}.$$

We want to approximate the eigenvalues of A_1 and A_2 using the Power Method (PM) and Rayleigh Quotient Iterations (RQI). Clearly, A_1 is a symmetric matrix, and A_2 is a non-symmetric matrix. The eigenvalue RQI finds depends on the initial vector x_0 , while the power method always finds the largest eigenvalue. We use the initial vector $x_0^T = (1, 0, -1)$ for both methods. The eigenvalue errors for the two methods are plotted in figure 1. In order to plot using semilog, a "least error" was set to 10^{-10} .

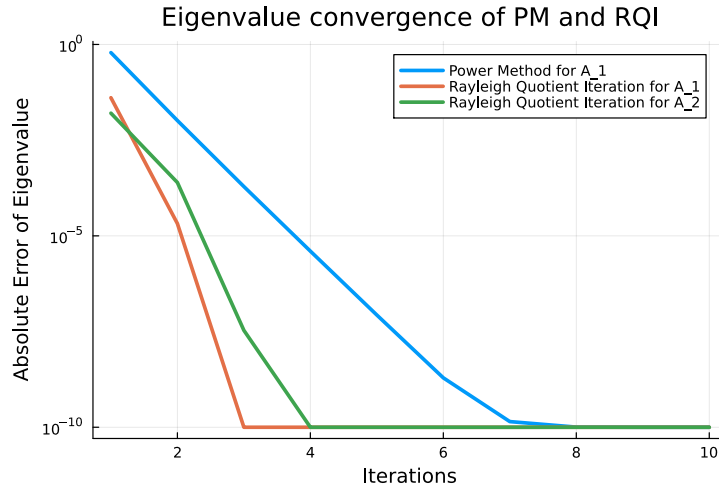


Figure 1: The absolute error of the approximated eigenvalue and the true value, using the Power Method for the matrix A_1 and Rayleigh Quotient Iteration for matrices A_1 and A_2 .

Let k be the number of iterations. The convergence factor of PM is determined by the two eigenvalues with largest absolute value. These are called λ_1 and λ_2 respectively. For A_1 , the convergence rate is thus found through

$$O\left(\left|\frac{\lambda_2}{\lambda_1}\right|^{2k}\right) = O\left(\left|\frac{1.571}{10.612}\right|^{2k}\right) = O\left(|0.148|^{2k}\right).$$

We know that for symmetric matrices, RQI has a cubic convergence rate, $O(\|v^k \pm x_j\|^3)$, while for non-symmetric matrices the convergence rate is quadratic,

$O(\|v^k \pm x_j\|^2)$. This explains why we obtain slower convergence for matrix A_2 than for matrix A_1 using RQI.

Problem 2

m	Single GS		Modified GS		Double GS		Triple GS	
	Time	Orth	Time	Orth	Time	Orth	Time	Orth
5	0.66	2.72×10^{-12}	0.42	2.4×10^{-12}	0.63	9.8×10^{-14}	1.39	9.8×10^{-14}
10	0.94	1.0×10^{-11}	1.14	7.8×10^{-12}	1.16	1.0×10^{-13}	1.51	1.0×10^{-13}
20	2.25	4.9×10^{-11}	3.70	3.3×10^{-11}	2.78	2.0×10^{-13}	3.70	2.0×10^{-13}
50	12.81	4.2×10^{-10}	15.55	1.8×10^{-10}	18.25	2.8×10^{-13}	23.39	2.8×10^{-13}
100	31.41	2.2×10^{-9}	52.99	5.1×10^{-10}	43.48	4.1×10^{-13}	49.34	4.1×10^{-13}

Table 1: Timing and Orthogonality for Different Gram-Schmidt Methods

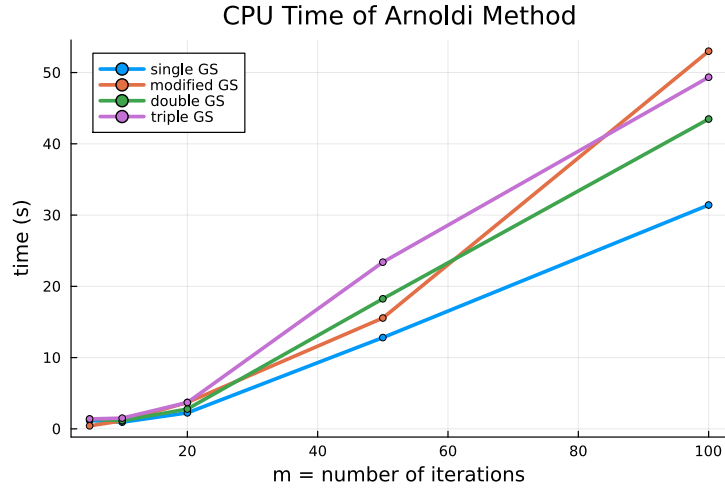


Figure 2: Arnoldi CPU times using different Gram Schmidt methods, for different number of iterations (m).

As shown in Figure 2, the modified Gram-Schmidt (GS) method is the slowest among the methods for large values of m . As expected, the triple GS method is slower than the double GS, while the single GS remains the fastest regardless of the number of iterations, as expected from the theory. However, we can see that the double and triple GS methods achieve better orthogonality, suggesting that they might be the "better" choices in this context. That said, since the orthogonality measure is effectively zero across all methods, one could argue that the faster methods are preferable for this specific problem. Furthermore, as shown in Table 1, the orthogonality indicator increases for each GS method with dimension m , which seems reasonable as we risk introducing further error with added dimensions of Q .

Problem 3

To prove that the approximation

$$\mu w = (K_m^T K_m)^{-1} K_m^T A K_m w, \text{ where } K_m = \left[\frac{b}{\|b\|}, \frac{Ab}{\|Ab\|}, \dots, \frac{A^{m-1}b}{\|A^{m-1}b\|} \right] \quad (1)$$

is equivalent to the Arnoldi's method for eigenvalue problems we shall inspect the equation further. Let the columns of K_m be represented by $k_1, \dots, k_m$. In Arnoldi's method one finds an orthogonal representation matrix, Q_m , of the Krylov subspace $\mathcal{K}_m(A, b) = \text{span}\{b, Ab, A^2b, \dots, A^{m-1}b\}$ and search for the eigenvalue such that $\mu w = H_m w = Q_m^T A Q_m w$, hence $\text{span}\{k_1, \dots, k_m\} = \text{span}\{q_1, \dots, q_m\}$. We will now show that (1) can be derived from the Arnoldi approximation.

Since the span of the columns of K_m and Q_m are equal there exists some invertible matrix $S \in \mathbb{R}^{m \times m}$ such that $K_m S = Q_m$. Thus, we can write the Arnoldi approximation as

$$\mu w = Q_m^T A Q_m w = S^T K_m^T A K_m S w. \quad (2)$$

Letting $u = S w$, and hence $w = S^{-1} u$, (2) becomes

$$\mu S^{-1} u = S^T K_m^T A K_m u. \quad (3)$$

Multiplying both sides of (3) by S^{-T} from the left, we get

$$\mu S^{-T} S^{-1} u = K_m^T A K_m u \quad (4)$$

Since, $I = Q_m^T Q_m = S^T K_m^T K_m S$, we see that $S^{-T} S^{-1} = K_m^T K_m$. Hence (4) can be written as

$$\mu K_m^T K_m u = K_m^T A K_m u. \quad (5)$$

Assuming $K_m^T K_m$ is invertible (5) is equivalent to

$$\mu u = (K_m^T K_m)^{-1} K_m^T A K_m u,$$

Thus, the approximation (1) is equivalent to the approximation generated by the Arnoldi method. $\square$

We now compare the approximations of the eigenvalues of A generated with the Arnoldi method, to the values found using (1). We run each method for m

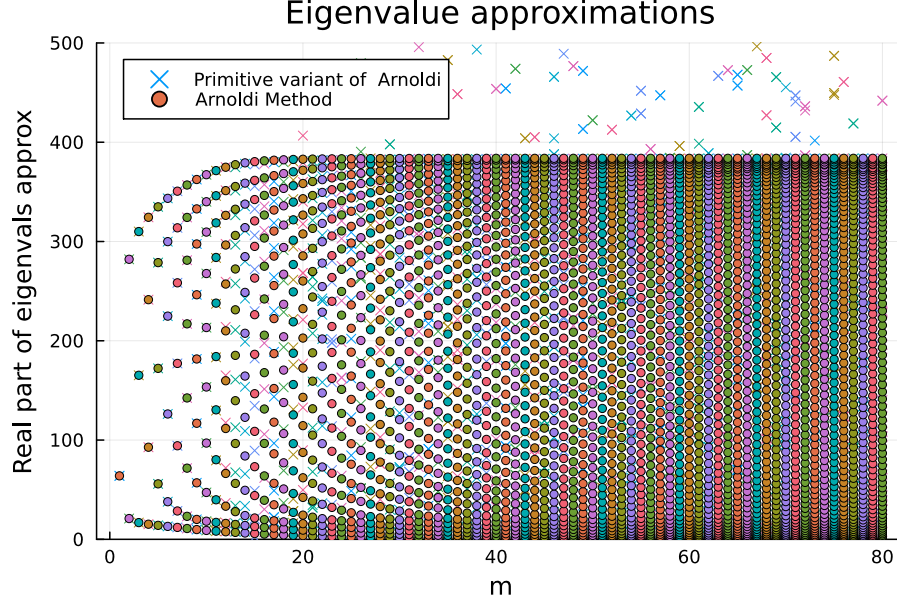


Figure 3: Eigenvalue approximations using the Arnoldi method and the primitive variant of the Arnoldi method.

iterations, letting m go from 1 to 80. For each m , both methods thus generate m eigenvalue approximations. We restrict the y -axis to $y \in [0, 500]$.

As we see in figure 3, the arnoldi method generates one more eigenvalue for each iteration, where all eigenvalues converge to some value between 0 and 400. Meanwhile, the primitive variant using the non-orthogonal basis seems to approximate eigenvalues that are further spread from those of the arnoldi method, the more iterations we compute. Had we not restricted the y -axis, we would see that some of the eigenvalues are very large negative/positive numbers when m is large. Already at $m = 15$ we can see that the two methods produce different approximations. This difference is due to the unorthogonality of K_m , which leads to larger errors the larger the matrix K_m becomes. Thus, we realize the importance of an orthogonal basis. Therefore, the Arnoldi method is much more stable, and hence the better method as m grows, which we need for good approximations.

Problem 4

In this part we look at eigenvalues of the given matrix Bwedge. We find the three eigenvalues that are the furthest from the rest at, approximately, $\lambda_1 = -47$, $\lambda_2 = 1 + 12i$ and $\lambda_3 = 1 - 12i$. This can be visually verified in the plot. The convergence factors are calculated by approximating the centers and radii of the respective discs, where disk i is the smallest circle that includes all eigenvalues except λ_i .

Eigenvalue	Disc Center, c	Disc Radius, ρ	Convergence Factor, $\alpha = \frac{\rho}{ c-\lambda }$
$\lambda_1 = -47$	$c_1 = -4.5$	$\rho_1 = 14$	$\alpha_1 = 0.33$
$\lambda_2 = 1 + 12i$	$c_2 = -24.5 + 13i$	$\rho_2 = 26$	$\alpha_2 = 0.72$
$\lambda_3 = 1 - 12i$	$c_3 = -24.5 - 13i$	$\rho_3 = 26$	$\alpha_3 = 0.72$

Table 2: Convergence Factors for Different Eigenvalues

We expect the fastest convergence from λ_1 , due to the small convergence factor.

In order to visualize the convergence of the eigenvalues for different numbers of iterations, we plot the true positions of λ_1, λ_2 and λ_3 in the complex plane, along with the positions of the approximated values for these three eigenvalues after each number of iterations. As we can see, λ_1 appears to be somewhat correctly approximated after only 3 iterations.

We also plot the error convergence of λ_1 , which has an error of 10^{-10} after about 13 iterations.

	Standard AM	$\sigma = -10$	$\sigma = -7 + 2i$	$\sigma = -9.8 + 1.5i$
$m = 10$	2.5	0.07	0.03	$1 \cdot 10^{-5}$
$m = 20$	0.6	$6 \cdot 10^{-4}$	$3 \cdot 4^{-8}$	$2 \cdot 10^{-14}$
$m = 30$	$2 \cdot 10^{-4}$	$3 \cdot 10^{-8}$	$6 \cdot 10^{-14}$	$1 \cdot 10^{-14}$

Table 3: Shifted Arnoldi method

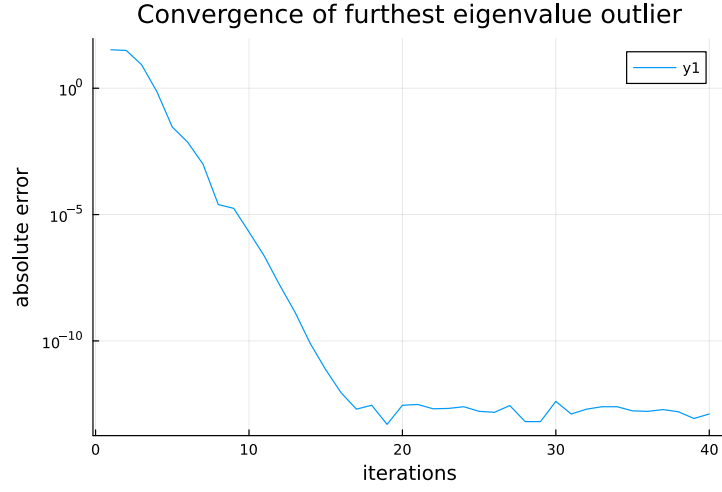


Figure 4: The convergence of the approximation of the fastest converging eigenvalue, λ_1 .

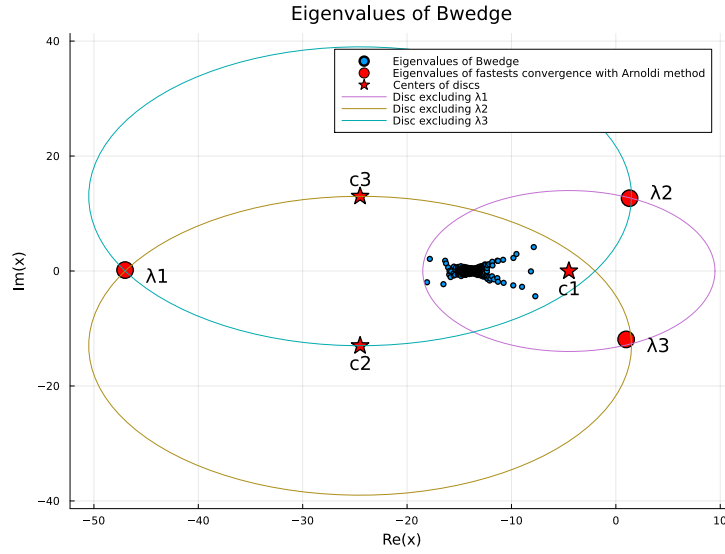


Figure 5: Eigenvalues of the "Bwedge" matrix, indicating the three eigenvalues for which we expect the fastest convergence. The circles with center c_i represent the discs that include all eigenvalues except λ_i .

As seen in figure 6 the arnoldi method for λ_1 is converging to its true value faster than the others. It is visually converging already when $m = 4$, and the three outliers have been approximated quite well when $m = 8$. For illustrative

purpose we add a plot for $m = 4, 5$ as well.

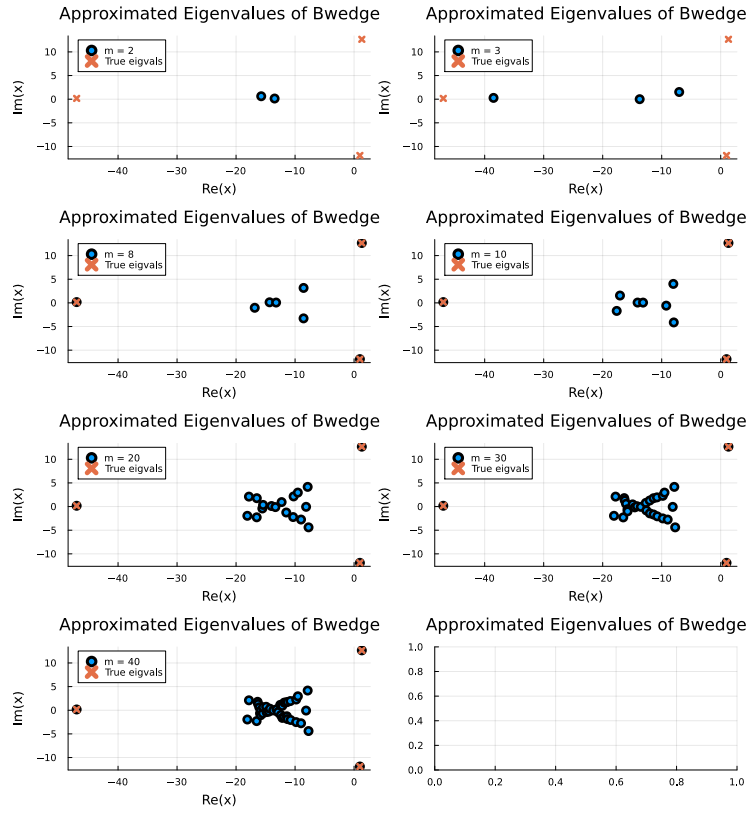


Figure 6: Approximations of the eigenvalues of "Bwedge" as the number of iterations increases.

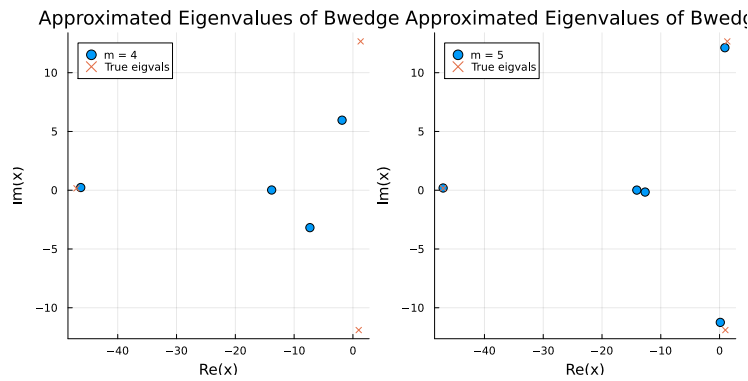


Figure 7: Approximations of the eigenvalues of "Bwedge" at iterations 4 and 5.

Problem 5

101: Block 1 applications

This video covers exploratory data analysis, which can be applied to areas like fraud detection, network analysis, and climate change.

Using this approach, a data set can be represented as a graph, with data points as nodes and similarities between points as edges. This graph is then represented by a Laplacian matrix, which thus captures the data points and their similarities. Since most data points are not very similar, this matrix is often sparse. Our goal is to approximate the eigenvalues of this large sparse matrix, as they can provide insights into the structure of the data.

One example is the MNIST data set which consists of hand written 3's and 8's. In this case, the eigenvalues of the corresponding Laplacian will give us insight into the clustering of the digits.

140: Rayleigh-Ritz method

The Rayleigh-Ritz method approximates eigenpairs of a matrix $A \in \mathbb{R}^{n \times n}$. Given an orthogonal matrix $Q \in \mathbb{R}^{n \times k}$, we use the span of Q 's columns as a search space for the eigenvector x , with $x \approx Qz$ for some $z \in \mathbb{R}^k$. If (x, λ) is an eigenpair of A , then $Ax = \lambda x$, which we approximate as $AQz \approx \lambda Qz$, or $AQz = \lambda Qz$. Using Q 's orthogonality, we get

$$Bz := Q^T AQz = \tilde{\lambda}z,$$

making $(\tilde{\lambda}, z)$ an eigenpair approximation for B , and $(\tilde{\lambda}, Qz)$ an eigenpair approximation for A . If x is in the span of Q 's columns, the Rayleigh-Ritz will give an exact approximation.

169: Breakdown

In the Gram-Schmidt orthogonalization within the Arnoldi method, we find $q_{m+1} = y/\beta$, where $\beta = h_{m+1,m}$ is the last element of the Hessenberg matrix. Breakdown of the Arnoldi method occurs when $\beta = 0$, as this leads to zero division.

In this video, we learn that Arnoldi breakdown will only happen when the eigenvalue approximation is exact. This follows directly from a lemma stating that the residual of the Ritz values is proportional to $h_{m+1,m} = \beta$. Thus, if $\beta = 0$, we will have zero residual, indicating that our Ritz values are exact.

168: Shift-and-invert Arnoldi method for eigenvalue problems

The Arnoldi method typically converges to extreme or isolated eigenvalues. To approximate other eigenvalues, we can apply the shift-and-invert Arnoldi method, which applies AM to the matrix $(A - \mu I)^{-1}$, where μ is a shift guides the convergence toward eigenvalues near μ . By modifying the Arnoldi method we can perform the shift-and-inverse without explicitly inverting the matrix.

Instead, we solve the linear system $w = (A - \mu I)Q(:, k)$, often through LU-factorization. After completing the shift-and-invert Arnoldi iterations, we extract the eigenvalue approximations of A near μ through $1.\backslash(eig(H[1 : m, 1 : m]) - \mu)$.